GROUP PROJECT

NUMERICAL AND SIMULATION TECHNIQUES IN FINANCE (FINM4053)

2025-11-23

Instructions

Form a team of 2 or 3 students and register it on iSpace. The goal is to write a Python software which follows the specifications detailed below. Submit your Python program on iSpace into one zipped file containing:

▷ code files (.py or .ipynb files)

▷ a final report containing your written answers, the user manual of your software, and the job allocation between the team members.

Each group only needs to submit one copy of the zip file. The deadline for the project will be released on iSpace.

Project description: pricing of barrier options by Monte Carlo simulations

Barrier options

Barrier options are a class of exotic financial options whose payoff depends on whether the underlying asset price touched a predefined barrier price during the lifetime of the option. Classical examples include:

Knock-out barrier options:

▷ **Up-and-out call option:** $P_T = (S_T - K)^+ \mathbf{1}\left\{\max_{0\le t\le T} S_t < B\right\}, \quad 0 < S_0 < B$

▷ **Down-and-out call option:** $P_T = (S_T - K)^+ \mathbf{1}\left\{\min_{0\le t\le T} S_t > B\right\}, \quad 0 < B < S_0$

▷ **Up-and-out put option:** $P_T = (K - S_T)^+ \mathbf{1}\left\{\max_{0\le t\le T} S_t < B\right\}, \quad 0 < S_0 < B$

▷ **Down-and-out put option:** $P_T = (K - S_T)^+ \mathbf{1}\left\{\min_{0\le t\le T} S_t > B\right\}, \quad 0 < B < S_0$

Knock-in barrier options:

▷ **Up-and-in call option:** $P_T = (S_T - K)^+ \mathbf{1}\left\{\max_{0\le t\le T} S_t \ge B\right\}, \quad 0 < S_0 < B$

▷ **Down-and-in call option:** $P_T = (S_T - K)^+ \mathbf{1}\left\{\min_{0\le t\le T} S_t \le B\right\}, \quad 0 < B < S_0$

▷ **Up-and-in put option:** $P_T = (K - S_T)^+ \mathbf{1}\left\{\max_{0\le t\le T} S_t \ge B\right\}, \quad 0 < S_0 < B$

▷ **Down-and-in put option:** $P_T = (K - S_T)^+ \mathbf{1}\left\{\min_{0\le t\le T} S_t \le B\right\}, \quad 0 < B < S_0$

Here $P_T \ge 0$ is the final cash-flow of the option, received at time T (the maturity of the option), $S_T \ge 0$ is the price of the underlying asset at time T, $K > 0$ is the strike price of the option, and $B > 0$ is the barrier price.

1. Monte Carlo pricing under Black-Scholes model

Suppose that the underlying asset price process follows a geometric Brownian motion with constant drift $r > 0$ and constant volatility $\sigma > 0$ under the risk-neutral pricing measure.

▷ Describe how to estimate the initial premiums of these eight barrier options by Monte Carlo simulations using the Euler scheme on a uniform time grid with N equal time intervals between $t = 0$ and the option maturity $t = T$.

▷ Implement these eight Monte Carlo estimators in Python. For your numerical tests, set $r = 0.05$, $\sigma = 0.20$, $T = 1$, $S_0 = K = 100$, and try a few different values of B.

▷ Sensitivity analysis with respect to the number M of Monte Carlo simulations: plot the estimated option premiums and 99% (asymptotic) confidence intervals with respect to M for each of these eight options (set B to an “interesting” value based on your previous tests).

▷ Sensitivity analysis with respect to the number N of time steps: fix M to a large value ensuring numerical convergence (based on your results from the above sensitivity analysis), then plot the estimated option premiums with respect to N for each of these eight options (set B to the same values as before).

2. Monte Carlo pricing under stochastic volatility model

Suppose now that the variance of the underlying asset price is not constant, but instead follows the 3/2 stochastic volatility model under the risk-neutral pricing measure, defined by

$$\begin{aligned} dS_t &= rS_t dt + \sqrt{V_t} S_t dW_t \\ dV_t &= \kappa(\theta V_t - V_t^2)dt + \lambda V_t^{3/2} dB_t \\ d\langle W, B\rangle_t &= \rho dt. \end{aligned}$$

In particular, the Brownian motions of the asset price and its variance are correlated, with correlation parameter $\rho \in (-1, 1)$.

▷ Redo all the questions from the previous section, with the constant volatility σ replaced by the 3/2 stochastic volatility model. For the numerical experiments, use the same numerical parameters as in the previous section, as well as $V_0 = \theta = 0.20$, $\kappa = 0.20$, $\lambda = 0.67$ and $\rho = -0.50$

▷ Sensitivity analysis with respect to the volatility-of-volatility: fix M to a large value ensuring numerical convergence, then plot the estimated option premiums with respect to λ for each of these eight options (set B to the same values as before). Conclude about the effect of stochastic volatility on the premiums of barrier options.

▷ Sensitivity analysis with respect to the correlation: fix M to a large value ensuring numerical convergence, then plot the estimated option premiums with respect to ρ for each of these eight options (set B to the same values as before). Conclude about the effect of the correlation between price and volatility on the premiums of barrier options.